Table C2. Fully unconstrained random parameters binary logit models with heterogeneity in means and variances of crash injury severity (serious injury/fatal vs. PDO/minor injury), estimated separately for each year: crashes involving a motorcycle and another vehicle type, Thailand, 2016–2023. Estimated parameter (*t*-statistic).

Variable description	2016	2017	2018	2019	2020	2021	2022	2023
Constant [serious injury / fatal]								
Constant	0.799 (2.27)	0.554 (1.65)	0.320 (0.97)	0.338 (0.93)	–0.539 (–1.92)	0.013 (0.09)	–0.669 (–2.63)	0.004 (0.02)
Vehicle involvement								
One motorcycle involved (1 if yes, 0 otherwise)	–0.868 (–2.65)	–0.608 (–1.90)	–0.603 (–2.02)	–0.522 (–1.67)	–0.689 (–2.61)			
One passenger car involved (1 if yes, 0 otherwise)	–0.555 (–4.69)		–0.645 (–5.55)	–0.389 (–2.20)		–0.663 (–6.03)	–0.498 (–4.62)	–1.012 (–6.00)
One pickup truck involved (1 if yes, 0 otherwise)				–0.402 (–1.92)	0.226 (1.73)			–0.376 (–2.17)
(Standard deviation of parameter distribution)				1.516 (3.71)	1.382 (4.22)			1.081 (2.57)
Two or more pickup trucks involved (1 if yes, 0 otherwise)				–0.899 (–2.26)	–0.500 (–1.65)		–1.050 (–2.85)	–0.585 (–1.88)
One 6-wheel truck involved (1 if yes, 0 otherwise)	0.942 (3.13)	1.158 (3.58)	1.500 (4.42)	0.550 (1.81)	1.450 (6.41)	0.486 (2.17)	0.815 (3.36)	
One 10-wheel truck involved (1 if yes, 0 otherwise)	1.280 (4.08)	1.146 (3.66)	1.183 (3.71)	1.087 (3.13)	2.008 (7.03)	0.734 (2.66)	1.195 (4.00)	0.758 (2.21)
One truck with more than 10 wheels involved (1 if yes, 0 otherwise)	1.161 (4.81)	1.785 (7.57)	1.401 (5.93)	1.283 (5.05)	1.738 (9.07)	1.127 (5.46)	1.392 (6.89)	1.081 (4.46)
Van involved (1 if yes, 0 otherwise)		0.888 (2.76)						–0.953 (–2.36)
Crash type								
Rear-end collision in the same lane (1 if yes, 0 otherwise)	–0.436 (–3.28)		–0.444 (–3.41)			–0.270 (–1.91)		
Sideswipe (parallel movement) (1 if yes, 0 otherwise)	–0.824 (–4.96)	–0.277 (–1.82)	–1.112 (–4.20)	–0.800 (–4.91)	–0.544 (–4.27)	–0.672 (–4.19)	–0.660 (–4.93)	–0.861 (–5.82)
(Standard deviation of parameter distribution)			2.010 (3.22)					
Head-on collision (1 if yes, 0 otherwise)	0.838 (1.90)	1.116 (3.75)	0.772 (3.69)	1.255 (4.28)	1.495 (6.12)	0.912 (4.07)	0.860 (3.48)	1.174 (4.95)
Crash cause								
Speeding (1 if yes, 0 otherwise)		–0.322 (–3.01)					0.491 (2.13)	
Cutting in too closely (1 if yes, 0 otherwise)	0.016 (0.08)		0.151 (0.93)		0.208 (1.63)	–0.166 (–1.09)	0.541 (2.24)	
(Standard deviation of parameter distribution)	2.394 (4.15)		2.200 (4.66)		1.333 (3.53)	1.339 (2.96)		
Running a red light (1 if yes, 0 otherwise)	1.229 (3.10)		1.072 (3.40)	0.742 (2.35)	1.007 (3.70)	0.515 (2.02)	0.968 (2.86)	
Roadway characteristics								
Crash on the main carriageway (1 if yes, 0 otherwise)		–0.568 (–2.80)				0.631 (2.25)		–0.630 (–2.22)
Crash on a frontage road (1 if yes, 0 otherwise)	–1.604 (–5.26)	–1.923 (–5.98)	–1.146 (–3.73)	–0.524 (–1.89)	–0.830 (–3.47)		–1.389 (–5.57)	–0.858 (–3.66)
Road under construction (1 if yes, 0 otherwise)								–0.818 (–2.73)
Painted median (1 if yes, 0 otherwise)							–0.612 (–3.46)	
Raised or earth median (1 if yes, 0 otherwise)	–0.740 (–5.39)	–0.334 (–2.62)	–0.382 (–2.63)	–0.383 (–2.80)		–0.318 (–2.53)	–0.425 (–3.66)	
Depressed median (1 if yes, 0 otherwise)		0.001 (0.00)	0.353 (2.15)	0.307 (2.01)	0.500 (3.82)	0.520 (2.86)		0.477 (3.27)
(Standard deviation of parameter distribution)		1.432 (2.43)				1.313 (2.50)		
Barrier median (1 if yes, 0 otherwise)	–1.425 (–6.53)	–1.171 (–6.09)	–0.811 (–3.93)	–1.285 (–5.53)	–0.691 (–3.87)	–0.740 (–3.80)	–1.035 (–5.60)	–0.829 (–4.35)
Horizontal curve (1 if yes, 0 otherwise)	0.688 (3.14)					0.413 (2.20)	0.106 (0.28)	
(Standard deviation of parameter distribution)							3.308 (2.34)	
Concrete pavement surface (1 if yes, 0 otherwise)			–0.740 (–4.29)		–0.548 (–3.96)	–0.665 (–4.60)		–0.862 (–2.56)
(Standard deviation of parameter distribution)								2.755 (3.13)
T-intersection (1 if yes, 0 otherwise)	–0.401 (–1.70)		0.379 (1.66)	–0.416 (–1.74)				
Four-leg intersection (1 if yes, 0 otherwise)			0.729 (2.77)				–0.419 (–1.99)	
U-turn without auxiliary lane (1 if yes, 0 otherwise)					0.614 (1.77)			

Variable description	2016	2017	2018	2019	2020	2021	2022	2023
U-turn with deceleration lane only (1 if yes, 0 otherwise)	0.984 (2.50)	0.567 (1.81)			0.749 (2.25)			
U-turn with deceleration and acceleration lanes (1 if yes, 0 otherwise)	0.937 (2.29)					1.024 (2.47)	0.731 (1.75)	
Environmental characteristics								
Wet road surface (1 if yes, 0 otherwise)								–1.081 (–2.12)
Raining (1 if yes, 0 otherwise)	1.051 (2.97)							1.057 (1.88)
Dark, no street lighting (1 if yes, 0 otherwise)	0.807 (4.14)	0.754 (4.00)	0.999 (4.75)	0.935 (4.06)	0.943 (4.38)	1.133 (5.06)	0.411 (1.95)	
Dark, with street lighting (1 if yes, 0 otherwise)		0.219 (1.81)	0.505 (3.96)		0.247 (1.88)			
Temporal characteristics								
Songkran festival period (1 if yes, 0 otherwise)		–0.442 (–3.16)	–0.346 (–2.17)	–0.638 (–2.29)		–0.572 (–3.20)	–0.323 (–2.14)	–0.359 (–2.14)
(Standard deviation of parameter distribution)				2.992 (2.76)				
New Year festival period (1 if yes, 0 otherwise)	–0.328 (–2.09)		–0.365 (–2.39)	–0.705 (–4.96)	–0.431 (–2.71)	–0.492 (–3.21)	–0.787 (–1.96)	–0.546 (–3.32)
(Standard deviation of parameter distribution)							2.834 (2.33)	
Public holiday (1 if yes, 0 otherwise)			0.707 (1.66)					0.594 (1.66)
Evening (1 if yes, 0 otherwise)							0.419 (3.29)	0.509 (4.22)
Midnight / early morning (1 if yes, 0 otherwise)	0.328 (1.77)	0.610 (3.30)	0.613 (3.32)	0.827 (4.52)	0.871 (4.69)	0.677 (3.31)	0.615 (3.56)	1.174 (6.19)
Peak hours (1 if yes, 0 otherwise)	–0.125 (–0.76)							
(Standard deviation of parameter distribution)	1.408 (3.05)							
Region								
Eastern region (1 if yes, 0 otherwise)	1.357 (5.59)	0.972 (4.51)	0.922 (4.18)	1.394 (5.80)	1.493 (6.34)	0.786 (3.76)	0.681 (3.78)	1.198 (5.36)
Northeastern region (1 if yes, 0 otherwise)	1.046 (6.64)	0.451 (3.61)	1.113 (6.91)	1.248 (7.67)	1.436 (9.14)	0.965 (6.35)	1.045 (7.27)	1.250 (7.87)
Northern region (1 if yes, 0 otherwise)	0.399 (2.13)		0.468 (2.34)	0.692 (3.59)	–0.317 (–2.16)	–0.581 (–3.92)	–0.350 (–2.14)	
Southern region (1 if yes, 0 otherwise)			0.668 (3.98)	0.905 (5.18)	0.710 (4.40)	0.530 (3.24)	0.595 (3.60)	0.592 (3.87)
Western region (1 if yes, 0 otherwise)	0.895 (3.71)	0.426 (2.35)	0.827 (3.38)	0.873 (3.82)	1.244 (5.56)	0.748 (3.25)	0.706 (3.44)	0.823 (3.83)
Heterogeneity in the means of random parameters								
One pickup truck involved: Evening				0.445 (1.84)	0.654 (2.79)			
One pickup truck involved: Peak hours				0.451 (1.96)				
Cutting in too closely: Two or more pickup trucks involved	–3.053 (–2.03)							
Cutting in too closely: Van involved			1.774 (1.94)					
Cutting in too closely: Depressed median	1.351 (2.63)							
Cutting in too closely: Raining						1.647 (1.79)		
Cutting in too closely: Public holiday					–1.570 (–1.96)			
Depressed median: U-turn without auxiliary lane		3.240 (2.18)						
Depressed median: Wet road surface						–1.829 (–2.82)		
Depressed median: Peak hours		–0.683 (–1.76)						
Horizontal curve: Public or commercial access point							3.898 (1.82)	
Concrete pavement surface: U-turn without auxiliary lane								3.072 (2.00)
Concrete pavement surface: Northern region								1.659 (1.89)
New Year festival period: Peak hours							–1.704 (–2.04)	
Model statistics								

Variable description	2016	2017	2018	2019	2020	2021	2022	2023
Number of observations	2,617	2,269	2,759	2,583	3,327	2,681	2,511	2,624
Number of estimated parameters	31	25	32	29	30	30	32	30
Random parameters / heterogeneity in means / in variances	2 / 2 / 0	1 / 2 / 0	2 / 1 / 0	2 / 2 / 0	2 / 2 / 0	2 / 2 / 0	2 / 2 / 0	2 / 2 / 0
Log-likelihood at zero, LL(0)	–1813.97	–1572.75	–1912.39	–1790.40	–2306.10	–1858.33	–1740.49	–1818.82
Log-likelihood with constant only, LL(C)	–1784.04	–1570.03	–1907.69	–1788.35	–2284.92	–1847.48	–1722.89	–1808.90
Log-likelihood at convergence, LL(β)	–1396.36	–1304.26	–1519.03	–1512.96	–1829.08	–1532.50	–1447.42	–1468.60
McFadden pseudo $\rho^2 = 1 - LL(\beta)/LL(0)$	0.230	0.171	0.206	0.155	0.207	0.175	0.168	0.193

Notes. A positive parameter increases the probability of a serious injury/fatal outcome. Reference categories: Central region, daytime, no median. Each column is a separate model for one year. Each model allows at most two random parameters, two heterogeneity-in-means terms and one heterogeneity-in-variance term, searched with 100 Halton draws at the 90% level and confirmed with 2,000 draws; non-significant parameters were then removed, so all parameters except constants and the means of random parameters are statistically significant at the 90% level. A blank cell means the variable is not in that year's model. Rows with a standard deviation give the mean of the random parameter in the years where it is random and a fixed parameter in the other years. A variable used in a heterogeneity function enters only the random parameter shown in that year and is removed from the main utility. Random parameters normally distributed; 2,000 Halton draws.